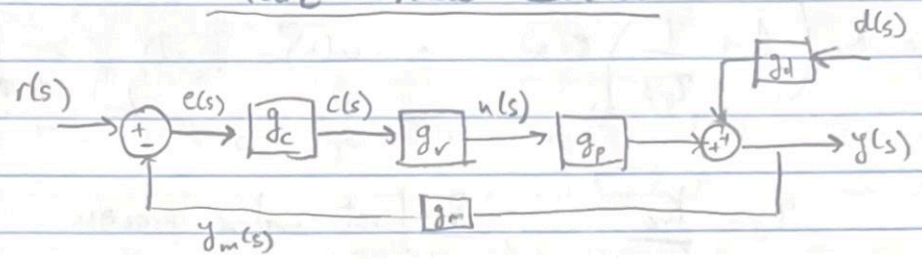
Controls

Quiz Cheat Sheet



$$G_{CL} = \frac{G_p G_v G_c}{1 + G_p G_v G_c G_m}$$

Valve and measurement are usually lumped w/ process transfer function

+ For first order process (w/ v, m and D ignored):

Proportional Control:

$$g_{CL}(s) = \frac{k_{CL}}{\tau_{CL} s + 1} \quad ; \quad g_c = K_c \quad ; \quad y(t) = M k_{CL}\left(1 - e^{-t/\tau_{CL}}\right)$$

$$g_p = \frac{k_p}{(\tau_p s + 1)}$$

derived for step change in

$$g_{CL}(s) = \frac{y}{y_{sp}}$$

Laplace transform this. ←

$$\frac{y}{y_{sp}} = \frac{k_{CL}}{\tau_{CL} s + 1}$$

$$k_{CL} = \frac{k_p k_c}{1 + k_p k_c} \quad ; \quad \tau_{CL} = \frac{\tau_p}{1 + k_p k_c}$$

Proportional will always have offset:

$y(t) \neq M$ since we have $(1 - \underbrace{e^{-t/(\tau_{CL})}}_{\text{never equals 0}})$

$\rightarrow K_c \propto \frac{1}{\tau_{CL}}$

$\rightarrow$ As $\uparrow K_c \rightarrow$ offset is reduced (for given k_p)

$$\text{offset} = \lim_{t \to \infty} M - y(t \to \infty) = M[1 - k_{CL}] = M\left[\frac{1}{1 + k_p k_c}\right]$$

PI control

$$u(s) = K_c\left(1 + \frac{1}{\tau_I s}\right)e(s) \Rightarrow u(t) = K_c e(t) + K_I \int_0^t e(\theta)\,d\theta$$

$$\tau_I = \frac{K_c}{K_I}$$; for first order process:

$$\frac{y}{y_{sp}} = \frac{\tau_I s + 1}{\frac{\tau_I \tau_p}{K_p K_c} s^2 + \frac{\tau_I (1 + K_p K_c)}{K_p K_c} s + 1}$$

→ $\tau \uparrow$ → "slower" process

→ If $\tau_I = \tau_p$ → first order response (where $\tau_{CL} = \frac{\tau_p}{K_p K_c}$)

PID Control

$$u(s) = K_c\left(1 + \frac{1}{\tau_I s} + \tau_D s\right)e(s) = K_c\left(\frac{\tau_D \tau_I s^2 + \tau_I s + 1}{\tau_I s}\right)e(s)$$

↳ $u(t) = K_c e(t) + K_I \int_0^t e(\theta)\,d(\theta) + K_D \frac{de(t)}{dt}$

$$\tau_D = \frac{K_D}{K_c}$$

Ideal derivative controller: 1) More sensitive to measurement noise
2) Derivative "spike" for step change

DS

$$\frac{y}{y_{sp}} = G_{CL} = \frac{G_c G}{1 + G_c G} \quad ; \quad G_c = \frac{1}{\tilde{G}}\left[\frac{\left(\frac{y}{y_{sp}}\right)_d}{1 - \left(\frac{y}{y_{sp}}\right)_d}\right]$$

For non-minimum: Have to include time-delay or RHP in G_c expression

IMC

$$\tilde{G} = \xi_+ \tilde{G}_- \quad ; \quad G_c^* = \frac{1}{\tilde{G}_-} f \quad ; \quad G_c = \frac{G_c^*}{1 - G_c^* \tilde{G}} \quad ; \quad f_n = \frac{1}{(\tau s + 1)^n}$$

Dynamic Modelling

Assumptions:
1) Well mixed (low viscosity)
2) Constant density

Mass

$$\frac{dM}{dt} = m_{in} - m_{out} + m_{gen} - m_{con} \quad ; \quad M = V \cdot \rho \qquad m = F \cdot \rho$$

For component:

x [=] C_A [=] moles / volume : $M = x_i V$, $m = F_i \cdot x_i$ (Mole balance)

x [=] ppm = 10^{-6} (by mass) → unitless (mass fraction) : $m = F \cdot x \cdot \rho$, $M = x \cdot \rho \cdot V$

Energy

$$\frac{d\bar{H}}{dt} = \bar{F}_i \rho \bar{H}_i - F_p \bar{H} + Q + W_s \quad ; \quad H = \rho V \bar{H}$$

Assumptions
1) Constant heat capacity

$$\bar{H}(T) = \int_{T_{ref}}^{T} c_p \, dT \Rightarrow \bar{H}(T) = c_p (T - T_{ref})$$

$$\frac{d(V \rho c_p (T - T_{ref}))}{dt} = F_i \rho c_p (T_i - T_{ref}) - F_p c_p (T - T_{ref}) + r \Delta H \cdot V + Q + W_s$$

$Q = -UA(T - T_{cw})$

2) KE, PE effects are neglected
3) Change in PV term is neglected
4) Negligible shaft work
5) $\Delta H_{rxn} = \Delta H^{\circ}_{rxn} = H_{rxn}(T)$

Linear diff eq: variables are multiplied by a constant
↳ Can just replace vars w/ deviation vars and rearrange to get the linear state space model

Non-linear diff eq:

$$\dot{x} = A\bar{x} + B\bar{u}$$
$$\dot{y} = C\bar{x} + D\bar{u}$$

$$\dot{x} = \frac{dx}{dt} = f(x_1, x_2, u) = \left.\frac{\partial f}{\partial x_1}\right|_{ss} \bar{x}_1 + \left.\frac{\partial f}{\partial x}\right|_{ss} \bar{x}_2 + \left.\frac{\partial f}{\partial u}\right|_{ss} \bar{u}$$

$$A = \begin{bmatrix} \partial f_1/\partial x_1 & \partial f_1/\partial x_2 \\ \partial f_2/\partial x_1 & \partial f_2/\partial x_2 \end{bmatrix} \quad ; \quad B = \begin{bmatrix} \partial f_1/\partial u_1 \\ \partial f_2/\partial u_2 \end{bmatrix}$$

$\bar{x} = x - x_{ss}$

$sY(s) = sG(s)U(s)$

Final Value Theorem: $\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s)$ → $sY(s)$ should not have strictly positive real parts (and at most 1 pole @ $s=0$)

Initial Value Theorem: $\lim_{t\to 0} y(t) = \lim_{s\to\infty} sY(s)$ → $y(t)$ has to be cont.

Transfer Function Model

SISO: 1) n order ODE (use state space model to linearize)
2) Laplace transform of both sides
3) Get transfer function $G(s) = \frac{Y(s)}{U(s)}$
4) Study zeros & poles (pole zero or time constant form)

For multiple inputs → Have to find transfer function of each input to the output (However, it doesn't explicitly have to be related to the output)

+**First order:**

$$\tau_p \frac{dy}{dt} + y = K_p u \longrightarrow \frac{Y(s)}{U(s)} = \frac{K_p}{\tau_p s + 1}$$

Step response: $y(t) = K_p M (1 - e^{-t/\tau_p})$

Impulse: $y(t) = \frac{P k_p}{\tau_p} e^{-t/\tau_p}$

★ If RO = 1 (output has non-zero slope initially)
RO > 1 (output has zero slope initially)

(Pg 170)

+**Second order:** (Non-numerator dynamics) → s-shaped graphs

$$\tau^2 \frac{d^2y}{dt^2} + 2\xi\tau \frac{dy}{dt} + y = ku(t) \longrightarrow \frac{Y(s)}{U(s)} = \frac{k}{\tau^2 s^2 + 2\xi\tau s + 1}$$

$p_1 = \frac{-\xi}{\tau} + \frac{\sqrt{\xi^2-1}}{\tau}$

$p_2 = \frac{-\xi}{\tau} - \frac{\sqrt{\xi^2-1}}{\tau}$

$\xi > 1$	Overdamped	Two real, distinct roots
$\xi = 1$	Critically damped	Two real, equal roots
$0 < \xi < 1$	Underdamped (oscillatory)	Two complex conjugate roots

+ Second Order (w/ numerator dynamics)

gain/time constant form →

$$G_1(s) = \frac{k_p(\tau_a s + 1)}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

Relative order = (Denom − Num) order

Pole zero form →

$$G_1(s) = \frac{k_{p_2}(s - z_1)}{(s - p_1)(s - p_2)}$$

$$k_{p_2} = \frac{k\tau_p}{\tau_1 \tau_2}, \quad p_1 = \frac{-1}{\tau_1}, \quad p_2 = \frac{-1}{\tau_2}, \quad z = \frac{-1}{\tau_n}$$

For a step response:

i) $\tau_a < 0$: Inverse response (positive zeros) → parallel first order w/ opposite gain

ii) $\tau_a > \max(\tau_1, \tau_2)$: overshoot

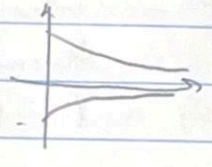
iii) $0 < \tau_a < \max(\tau_1, \tau_2)$: first order behaviour

+ Lead Lag:

$$G_1(s) = k_p \frac{\tau_a s + 1}{\tau_p s + 1}$$

↑ Don't forget to express in this form before calculations

Discont. response @ t=0



Step change:

i) $0 < \tau_1 \leq \tau_a$ → overshoot

ii) $0 < \tau_a \leq \tau_1$ → first order

iii) $\tau_a < 0$ → inverse response

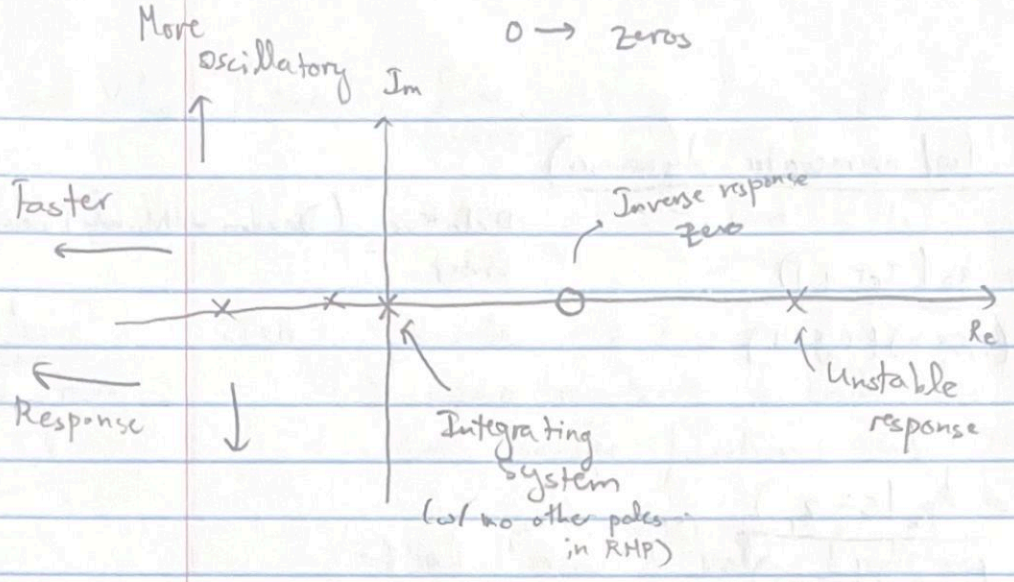
+ Dead time

Time delay of θ gives $e^{-\theta s}$ as transfer function

$$e^{-\theta s} = \frac{-\frac{\theta}{2}s + 1}{\frac{\theta}{2}s + 1} = \frac{+\frac{\theta^2}{12}s^2 - \frac{\theta}{2}s + 1}{\frac{\theta^2}{12}s^2 + \frac{\theta}{2}s + 1}$$

X → Poles

O → Zeros



Stable response

Trends & facts:

1) Multiple RHP zeros cause multiple changes in direction

↓

observed in dead time when second order app

Euler:

$\sin x = \frac{1}{2}(e^{ix} - e^{-ix})$ → Img

$\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$ → Real

Laplace transform $sX(s) = AX(s) + Bu(s)$; $Y(s) = CX(s)$

$$y(s) = \underbrace{[C(sI - A)^{-1}B + D]}_{G(s)} u(s)$$

No. of stat vars (n) = Order of denominator (i.e. n poles)

Going straight from state space → To transfer function model

Laplace $\mathcal{L}\{ae^{-bt}\} = \frac{a}{a+b}$ $F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st}\cdot dt = \int_0^\infty e^{-t(a+b+s)}\cdot dt$

$= \frac{1}{a+b+s}$

$\mathcal{L}\left\{\frac{df}{dt}\right\} = sF(s) - f(0)$

$\mathcal{L}\left\{\frac{df^2}{dt^2}\right\} = s^2F(s) - sf(0) - f'(0)$; Repeated roots: $\mathcal{L}^{-1}\left\{\frac{1}{(s+a)^3}\right\} = t^2e^{-at}$ ★